RB-101: Disk Space Exhaustion on Log Servers

Field	Value
Service	log-aggregator (hosts logsrv-01 .. logsrv-04)
Severity	SEV-2
Owner	Platform Ops team
Last reviewed	2026-09-14

1. Symptoms

- Alert **DiskUsageHigh** fires for mount point `/var/log`.
- Log shipping lag grows; Kibana shows gaps in the last 15 minutes.
- Applications log `No space left on device (errno 28)`.

2. Alert thresholds

Mount	Warning	Critical	Action
/var/log	85%	95%	Follow this runbook
/	80%	90%	Escalate to L2
/data	90%	97%	Expand volume (RB-120)

3. Response flow

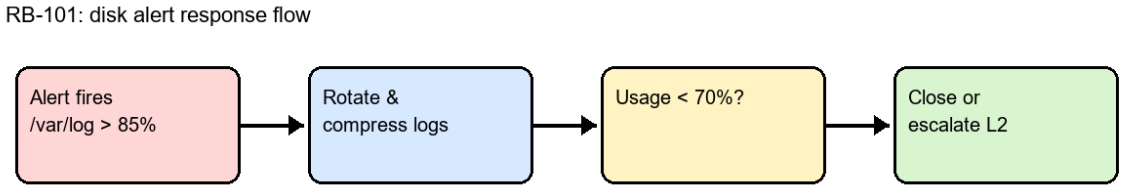

Figure 1: disk alert response flow.

4. Diagnosis

4.1 Find what is using the space

```
df -h /var/log
sudo du -xh /var/log | sort -rh | head -20
```

4.2 Check for deleted files still held open

A process can keep a deleted file open, so `du` and `df` disagree. Look for them with:

```
sudo lsof +L1 | grep /var/log
```

5. Remediation

1. Force a log rotation: `sudo logrotate -f /etc/logrotate.d/app`.
2. Compress logs older than 2 days: `find /var/log/app -name '*.log' -mtime +2 -exec gzip {} \;`
3. If a deleted file is held open, restart the owning service: `sudo systemctl restart filebeat`.
4. Confirm usage is below **70%** with `df -h /var/log`.

Warning: never delete files from /var/log/audit; they are retained for compliance (90 days).

6. Escalation

Level	Who	When	Contact
L1	On-call SRE	Immediately	#ops-oncall
L2	Platform Ops lead	Usage still > 90% after 30 min	PagerDuty: platform-ops
L3	Storage team	Volume must be expanded	PagerDuty: storage

7. Verification

- DiskUsageHigh alert resolves within 10 minutes.
- Log shipping lag back under 60 seconds.
- Ticket updated with root cause and the commands that were run.

8. Related runbooks

- RB-120: Expanding an LVM volume
- RB-415: High API latency